ANX-PR/CL/001-01
LEARNING GUIDE

SUBJECT

53002014 - Temas Avanzados De IngenierÍa TÉrmica

DEGREE PROGRAMME

05BK - Máster Universitario En Ingeniería De La Energía

ACADEMIC YEAR & SEMESTER

2025/26 - Semester 1

Index

Learning guide

1. Description

1.1. Subject details

Name of the subject	53002014 - Temas Avanzados de Ingeniería Térmica
No of credits	3 ECTS
Type	Optional/elective
Academic year ot the programme	First year
Semester of tuition	Semester 1
Tuition period	September-January
Tuition languages	English
Degree programme	05BK - Máster Universitario en Ingeniería de la Energía
Centre	05 - E.T.S. De Ingenieros Industriales
Academic year	2025-26

2. Faculty

2.1. Faculty members with subject teaching role

Name and surname	Office/Room	Email	Tutoring hours *
Luis Francisco Gonzalez Portillo (Subject coordinator)		lf.gonzalez@upm.es	Sin horario. Contactar con el profesor
Javier Muñoz Anton	Despacho	javier.munoz.anton@upm.es	Sin horario. Contactar con el profesor

Alberto Abanades Velasco	Despacho	alberto.abanades@upm.es	Sin horario. Contactar con el profesor
Victor Manuel Dominguez Perez	Despacho	victor.dominguez.perez@upm.es	Sin horario. Contactar con el profesor

* The tutoring schedule is indicative and subject to possible changes. Please check tutoring times with the faculty member in charge.

3. Prior knowledge recommended to take the subject

3.1. Recommended (passed) subjects

The subject - recommended (passed), are not defined.

3.2. Other recommended learning outcomes

- Industrial Heating and Cooling

- Thermal Power Plants

- Heat Transfer

- Thermal Engineering

4. Skills and learning outcomes *

4.1. Skills to be learned

CB10 - Que los estudiantes posean las habilidades de aprendizaje que les permitan continuar estudiando de un modo que habrá de ser en gran medida autodirigido o autónomo.

CB7 - Que los estudiantes sepan aplicar los conocimientos adquiridos y su capacidad de resolución de problemas en entornos nuevos o poco conocidos dentro de contextos más amplios (o multidisciplinares) relacionados con su área de estudio

CE16 - Aplicar conocimientos y habilidades adquiridas para la práctica profesional de alto nivel y la gestión de equipos en las empresas del sector energético.

CE2 - Analizar y establecer criterios de mejora energética y económica en instalaciones de generación y de consumo, incluyendo el sector transportes, conducente al diseño de alternativas más eficientes y con menor impacto ambiental.

CE8 - Disponer de habilidades, criterios y conocimientos para investigar, desarrollar e innovar en el campo de la energía: tecnologías renovables y no renovables, almacenamiento, vectores energéticos, en un contexto de decarbonización del sistema.

CE9 - Disponer de criterios y herramientas para entender la composición y características de los diferentes tipos de combustibles convencionales y no convencionales.

CG1 - Aplicar conocimientos de ciencias y tecnologías avanzadas a la práctica profesional o investigadora de la Ingeniería Energética.

CG2 - Poseer capacidad para diseñar, desarrollar, implementar, gestionar y mejorar productos, sistemas y procesos en los distintos ámbitos energéticos, usando técnicas analíticas, computacionales o experimentales avanzadas.

CT1 - Aplica. Habilidad para aplicar conocimientos científicos, matemáticos y tecnológicos en sistemas relacionados con la práctica de la ingeniería.

CT10 - Conoce. Conocimiento de los temas contemporáneos.

CT11 - Usa herramientas. Habilidad para usar las técnicas, destrezas y herramientas ingenieriles modernas necesarias para la práctica de la ingeniería.

CT12 - Es bilingüe. Capacidad de trabajar en un entorno bilingüe (inglés/español).

CT3 - Diseña. Habilidad para diseñar un sistema, componente o proceso que alcance los requisitos deseados teniendo en cuenta restricciones realistas tales como las económicas, medioambientales, sociales, políticas, éticas, de salud y seguridad, de fabricación y de sostenibilidad.

CT5 - Resuelve. Habilidad para identificar, formular y resolver problemas de ingeniería.

4.2. Learning outcomes

RA106 - Calcular los parámetros y los caudales de aire y combustible de procesos de combustión y postratamiento de gases utilizados en la industria y el trasporte

RA107 - Explicar las partes de una central térmica y la influencia que los distintos parámetros que las definen tienen en la eficiencia del proceso

RA104 - Identificar cómo mejorar la eficiencia de un ciclo para producción de frío y calor y sus consecuencias económicas y ambientales a través de modelos matemáticos

RA108 - Explicar un trabajo relacionado con la ingeniería energética mediante una presentación oral y un informe escrito

RA105 - Explicar y evaluar las características, fortalezas y debilidades, y alternativas de vectores energéticos disponibles en un contexto de descarbonización del sistema energético

* The Learning Guides should reflect the Skills and Learning Outcomes in the same way as indicated in the Degree Verification Memory. For this reason, they have not been translated into English and appear in Spanish.

5. Brief description of the subject and syllabus

5.1. Brief description of the subject

The course aims to deepen knowledge in Thermal Engineering by building upon concepts learned during the undergraduate degree. To this end, it reviews the current and future landscape of thermal energy and adopts a practical approach to the energy analysis of various thermal systems: thermal power plants, systems at ultra-high temperature, heating and cooling systems, industrial combustion applications, and technologies for reducing environmental impact. In addition, the course explores different potential energy vectors in decarbonization scenarios.

5.2. Syllabus

1. Thermal Energy: Present Context and Outlook
2. Advanced Thermal Power Plants
3. Industrial Applications of Combustion
4. Technologies for Reducing Environmental Impact
5. Heating and Cooling Production Systems
6. Energy Vectors for a Decarbonized Future

6. Schedule

6.1. Subject schedule*

Week	Type 1 activities	Type 2 activities	Distant / On-line	Assessment activities
1	**Overview of Thermal Energy: Past, Present, and Future of Thermal Energy and the Importance of Thermal Power Plants in the Electric Power System** Duration: 01:00 Lecture **Advanced Thermal Power Plants: Overview of the Main Types of Thermal Power Plants** Duration: 01:00 Lecture			
2	**Advanced Thermal Power Plants: Solar Thermal and Nuclear Power Plants, Hybridization, and Thermal Energy Storage** Duration: 01:00 Lecture **Advanced Thermal Power Plants: Next-Generation Thermal Power Plants** Duration: 01:00 Lecture			
3	**Advanced Thermal Power Plants: Application of Specialized Simulation Software** Duration: 02:00 Problem-solving class			
4	**Advanced Thermal Systems: Ultra-High Temperature** Duration: 02:00 Cooperative activities			**Project: Design and Analysis of an Advanced Thermal System** Group work Progressive assessment and Global Examination Not Presential Duration: 00:00
5	**Industrial Applications of Combustion: Fundamentals of Combustion** Duration: 01:00 Lecture **Industrial Applications of Combustion: Combustion of Gases and Liquids** Duration: 01:00 Lecture			

6	**Industrial Applications of Combustion: Solid Fuel Combustion** Duration: 01:00 Lecture **Industrial Applications of Combustion: Sizing and Design of a Combustion System for an Industrial Application Selected by the Students (e.g., burners, furnaces, turbines, engines, etc.)** Duration: 01:00 Problem-solving class			
7	**Technologies for Reducing Environmental Impact: Formation of Pollutant Emissions** Duration: 01:00 Lecture **Technologies for Reducing Environmental Impact: Emission Reduction Techniques** Duration: 01:00 Lecture			
8	**Technologies for Reducing Environmental Impact: Gas Post-Treatment Systems** Duration: 01:00 Lecture **Technologies for Reducing Environmental Impact: Emission Calculation and Design of a Gas Post-Treatment System for a Student-Selected Installation** Duration: 01:00 Problem-solving class			**Project: Combustion Systems and Their Environmental Impact** Group work Progressive assessment and Global Examination Not Presential Duration: 00:00
9	**Heating and Cooling Production Systems: Introduction to Heating and Cooling Systems** Duration: 01:00 Lecture **Heating and Cooling Production Systems: Fundamentals of Refrigeration** Duration: 01:00 Lecture			
10	**Heating and Cooling Production Systems: Heat Pumps** Duration: 01:00 Problem-solving class **Heating and Cooling Production Systems: Evaluation of Heating and Cooling Generation Systems** Duration: 01:00 Problem-solving class			**Project: Heating and Cooling Production Systems** Group work Progressive assessment and Global Examination Not Presential Duration: 00:00

11	**Energy Vectors in Decarbonization Scenarios: Understanding the Power-to-X Concept** Duration: 01:00 Lecture **Energy Vectors in Decarbonization Scenarios: Energy Vectors such as Synthetic Natural Gas, Ammonia, Methanol, and Hydrogen** Duration: 01:00 Lecture			
12	**Energy Vectors in Decarbonization Scenarios: Conversion and Interconnection Between Energy Vectors** Duration: 02:00 Lecture			**Project: Energy Vectors** Group work Progressive assessment and Global Examination Not Presential Duration: 00:00
13	**Presentation of Projects and Discussion** Duration: 02:00 Cooperative activities			
14	**Presentation of Projects and Discussion** Duration: 02:00 Cooperative activities			**Presentation of Projects Chosen by the Students** Group presentation Progressive assessment and Global Examination Presential Duration: 00:00
15				**Class Participation** Other assessment Progressive assessment Presential Duration: 00:00
16				
17				**Exam** Written test Global examination Presential Duration: 00:00

Depending on the programme study plan, total values will be calculated according to the ECTS credit unit as 26/27 hours of student face-to-face contact and independent study time.

7. Activities and assessment criteria

7.1. Assessment activities

7.1.1. Assessment

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
4	Project: Design and Analysis of an Advanced Thermal System	Group work	No Presential	00:00	15%	3 / 10	CB10 CT10 CE2 CB7 CT1 CT5 CT11 CE16 CT3 CE8 CG1 CG2 CT12
8	Project: Combustion Systems and Their Environmental Impact	Group work	No Presential	00:00	15%	3 / 10	CB10 CT10 CE2 CE9 CB7 CT1 CT5 CT11 CE16 CT3 CE8 CG1 CG2
10	Project: Heating and Cooling Production Systems	Group work	No Presential	00:00	15%	3 / 10	CB10 CT10 CE2 CB7 CT1 CT5 CT11 CE16 CT3 CE8 CG1 CG2

12	Project: Energy Vectors	Group work	No Presential	00:00	15%	3 / 10	CB10 CT10 CE2 CE9 CB7 CT1 CT5 CT11 CE16 CT3 CE8 CG1 CG2 CT12
14	Presentation of Projects Chosen by the Students	Group presentation	Face-to-face	00:00	15%	/ 10	CB7 CE16
15	Class Participation	Other assessment	Face-to-face	00:00	25%	3 / 10	CT10 CE16 CE8

7.1.2. Global examination

Week	Description	Modality	Type	Duration	Weight	Minimum grade	Evaluated skills
4	Project: Design and Analysis of an Advanced Thermal System	Group work	No Presential	00:00	15%	3 / 10	CB10 CT10 CE2 CB7 CT1 CT5 CT11 CE16 CT3 CE8 CG1 CG2 CT12
8	Project: Combustion Systems and Their Environmental Impact	Group work	No Presential	00:00	15%	3 / 10	CB10 CT10 CE2 CE9 CB7 CT1 CT5 CT11 CE16 CT3 CE8 CG1 CG2

10	Project: Heating and Cooling Production Systems	Group work	No Presential	00:00	15%	3 / 10	CB10 CT10 CE2 CB7 CT1 CT5 CT11 CE16 CT3 CE8 CG1 CG2
12	Project: Energy Vectors	Group work	No Presential	00:00	15%	3 / 10	CB10 CT10 CE2 CE9 CB7 CT1 CT5 CT11 CE16 CT3 CE8 CG1 CG2 CT12
14	Presentation of Projects Chosen by the Students	Group presentation	Face-to-face	00:00	15%	/ 10	CB7 CE16
17	Exam	Written test	Face-to-face	00:00	25%	3 / 10	

7.1.3. Referred (re-sit) examination

No se ha definido la evaluación extraordinaria.

7.2. Assessment criteria

The evaluation is primarily based on the results of projects carried out by the students, preferably in groups*, throughout the course. These projects cover the various topics of the subject (advanced thermal power systems, heating and cooling production systems, industrial applications of combustion, technologies for reducing environmental impact, and energy vectors in decarbonization scenarios) with a practical approach, while also incorporating the theoretical context. Each project is evaluated separately through a written report submitted by the group*, with each project contributing 15% to the final grade. Completion of all projects is mandatory (minimum grade: 3/10) in order to meet all course competencies. Optionally, students may give an oral presentation of any of the projects to earn an additional 15% of the final grade. Therefore, project-related work (including the optional oral presentation) represents 75% of the total grade. The remaining 25% can be obtained either through class participation (Kahoots, questions, etc.) during continuous assessment or through a final exam in the global assessment phase. In both cases, a minimum grade of 3/10 is required to pass the course.

* By default, the project must be carried out in groups. Only in justified cases will individual work be allowed.

8. Teaching resources

8.1. Teaching resources for the subject

Name	Type	Notes
Refrigeration and Air Conditioning	Bibliography	W.F. Stoecker, ?Refrigeration and Air Conditioning?, McGrawHill, 1982
Introduction to Refrigeration and Air Conditioning Systems Paperback	Bibliography	Allan Kirkpatrick, Introduction to Refrigeration and Air Conditioning Systems Paperback, Morgan & Claypool Publishers,2017
Refrigeration Systems and Applications	Bibliography	Ibrahim Dinçer and Mehmet Kanoglu, Refrigeration Systems and Applications, John Wiley and Sons, 2010

Advanced thermodynamics engineering	Bibliography	Annamalai, K., Puri, I.K., Jog, M.A., "Advanced thermodynamics engineering", CRC Press, 2011
Heat and mass transfer. A practical approach	Bibliography	Çengel, Y.A, "Heat and mass transfer. A practical approach", McGraw-Hill, 2007
Course notes	Web resource	Class notes (texts and slides) for the course uploaded to Moodle
Combustion Engineering	Bibliography	?Combustion Engineering? de K.W. Ragland y K. M. Bryden. Editorial: CRC Press y ?Fundamentals of Air Pollution Engineering? fr R.C. Flagan y J.H Sienfeld. Editorial: Prentice Hall.

9. Other information

9.1. Other information about the subject

The course is related to SDG 7, SDG 9, and SDG 11.